

## PH 503: Laser and Applications

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| Course Code   | PH 503                                                                  |
| Course Name   | Laser and Applications                                                  |
| L-T-P-C       | 3-0-0-3                                                                 |
| Prerequisites | Mechanics of Particles and Waves & Electrodynamics                      |
| Intended for  | B.Tech                                                                  |
| Distribution  | elective                                                                |
| Approval      | 2nd Adhoc Senate (May 2012), in the Minor in Applied Physics curriculum |

### Course Contents

- **Radiation**
- Energy density and pressure of radiation, cavity radiation, modes of oscillation. [1 Lectures]
- **Interaction of radiation with matter**
- Absorption, spontaneous and stimulated emission, Einstein coefficients, photoexcitation cross-section, amplification of radiation, laser pumping systems: optical pumping, electrical pumping other methods of pumping, spectral lines shapes, different types of broadening mechanism, gain calculation, threshold condition. [7 Lectures]
- **Cavity resonator**
- Time constant and quality factor of optical cavity, stability of resonators, g parameters, various types of resonators. [6 Lectures]
- Various Lasers: [8 Lectures]
  - Solid state lasers: Ruby Laser and Nd: YAG laser
  - Gas lasers: He-Ne laser, CO<sub>2</sub> laser and Nitrogen laser
  - Liquid lasers: Dye lasers
  - Semiconductor lasers
  - Free electron lasers
- **Laser pulse generation**
- Q-switching theory and various methods; mode locking: methods of mode locking, efficiency of mode locking, ultrashort (nanosecond, picosecond and femtosecond) laser pulse generation. [6 Lectures]
- **Applications in time-resolved spectroscopy**
- Fluorescence lifetime, various measurement techniques- oscilloscope method, time-correlated Application in optical communication Optical fibre, fibre laser. [2 Lectures]
- **Higher harmonic generation**
- White light continuum generation, optical parametric amplifier, pump-probe spectroscopy. [3 Lectures]
- **Holography**
- Theory, classification and application. [3 Lectures]

### Text Books

1. O. Svelto, Principles of lasers.
2. W. Koechner, Solid State Laser Engineering

### References

1. W. T. Silfvast, **Laser and Fundamentals**
2. A. E. Seigman, **Lasers**.
3. A. Yariv, **Quantum Electronics**.
4. D.R.Hall and P.E.Jackson (eds.), **The Physics and Technology of Laser Resonators**.
5. M.Young, **Optics and Lasers**.
6. D. Meschede, **Optics, Lights and Lasers**.
7. B.A.Lengyel, **Lasers**.